

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

**College Name: K.S.K College Of Engineering And
Technology College Code:8210**

Team ID: SWTID-2026-5917

S.No	Student Name	Register Number	NM ID
1	R. Keerthana(TL)	821023243014	4D6B65939BE956F8B63428DB37B6A6C8
2	A. Anitha	821023243003	7E772B7F47504B884AF49CECA7271713
3	K. Mahalakshmi	821023243015	E2856A855202337828544ACBF435E42E
4	P.M. Surekha	821023243028	D4D273F77C7294BA8EE4489C1E6BC14F

TABLE OF CONTENTS:

1. INTRODUCTION

1.1 Project Overview

1.2 Purpose

2. IDEATION PHASE

2.1 Problem Statement

2.2 Empathy Map Canvas

2.3 Brainstorming and Idea Selection

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

3.2 Functional Requirements

3.3 Non-Functional Requirements

3.4 Data Flow

3.5 Technology Stack

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

4.2 Proposed Solution

4.3 Solution Architecture

4.4 Data Model Design

4.5 Priority Decision Logic

5. PROJECT PLANNING & SCHEDULING

5.1 Agile Planning

5.2 Product Backlog

6. DEVELOPMENT PHASE

6.1 Salesforce Environment

6.2 Custom Object

7.FUNCTIONAL & PERFORMANCE TESTING

8.ADVANTAGES & DISADVANTAGES

9.CONCLUSION

10.FUTURE SCOPE

11.APPENDIX

1. INTRODUCTION

1.1 Project Overview :

Customer support centers often struggle with overwhelming ticket volumes, where manual prioritization leads to delays and inconsistent service. This project proposes a Salesforce-based solution integrated with Agentforce AI to intelligently analyze ticket descriptions, classify urgency, and automate assignment. By embedding flows and AI subagents, the system ensures that critical cases are escalated immediately, while routine issues are handled efficiently.

1.2 Purpose :

- Automate classification of tickets into High, Medium, or Low urgency.
- Link tickets with customer Accounts for contextual resolution.
- Reduce dependency on manual prioritization and assignment.
- Automatically generate urgent tasks for high-priority cases.
- Use Agentforce AI for NLP-based ticket analysis and backend automation.
- Improve operational efficiency, SLA compliance, and customer satisfaction.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	Role	Goal	Challenge	Reason	Feeling
PS-1	Support Agent	Focus on urgent tickets	Prioritization is manual	Large ticket queues	Stressed

PS-2	Customer	Quick resolution	Urgent cases delayed	High ticket volume	Anxious
PS-3	Manager	Assign tickets correctly	Manual judgment required	No consistent urgency signals	Pressure d
PS-4	Agent	Detect urgency	Keywords missed	Varied language in descriptions	Uncertain
PS-5	Manager	Reduce SLA risk	Old tickets overlooked	SLA monitoring inconsistent	Worried

2.2 Empathy Map Canvas

Support Agent Persona

- **Think & Feel:** Wants urgent issues flagged instantly.
- **Hear:** Customers demanding faster updates.
- **See:** Ticket queues, SLA deadlines, workload imbalance.
- **Say & Do:** Reviews tickets, escalates urgent cases, updates status.
- **Pain:** Manual prioritization, missed urgency signals.
- **Gain:** Automated classification, urgent task creation, reduced workload.

Customer Persona

- **Say:** “Why is my ticket still open?”
- **Think:** “Will my issue be resolved soon?”
- **Pain:** Long resolution times, lack of visibility.
- **Gain:** Faster resolution, real-time updates, improved trust.



2.3 Brainstorming & Idea Prioritization

Ideas Considered

- Manual registers
- Spreadsheet tracking
- Standalone portal
- Salesforce CRM + Agentforce

Final Choice: Salesforce CRM with Agentforce due to automation, AI-assisted analysis, and scalability.



Top Prioritized Features

- AI-based priority classification
- Automatic ticket assignment
- Real-time notifications
- SLA breach alerts
- Dashboards for performance monitoring

3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Ticket Creation → 2. Account Identification → 3. AI Analysis → 4. Auto Assignment → 5. Agent Resolution → 6. Notifications → 7. Resolution → 8. Feedback

Outcome: Faster resolution, informed customers, SLA compliance, and continuous improvement.



3.2 Functional Requirements

- FR-1: Ticket creation and management
- FR-2: Account association
- FR-3: Description analysis
- FR-4: Priority classification

- FR-5: Urgent task creation
- FR-6: Agent assignment
- FR-7: SLA risk monitoring
- FR-8: Conversational AI support

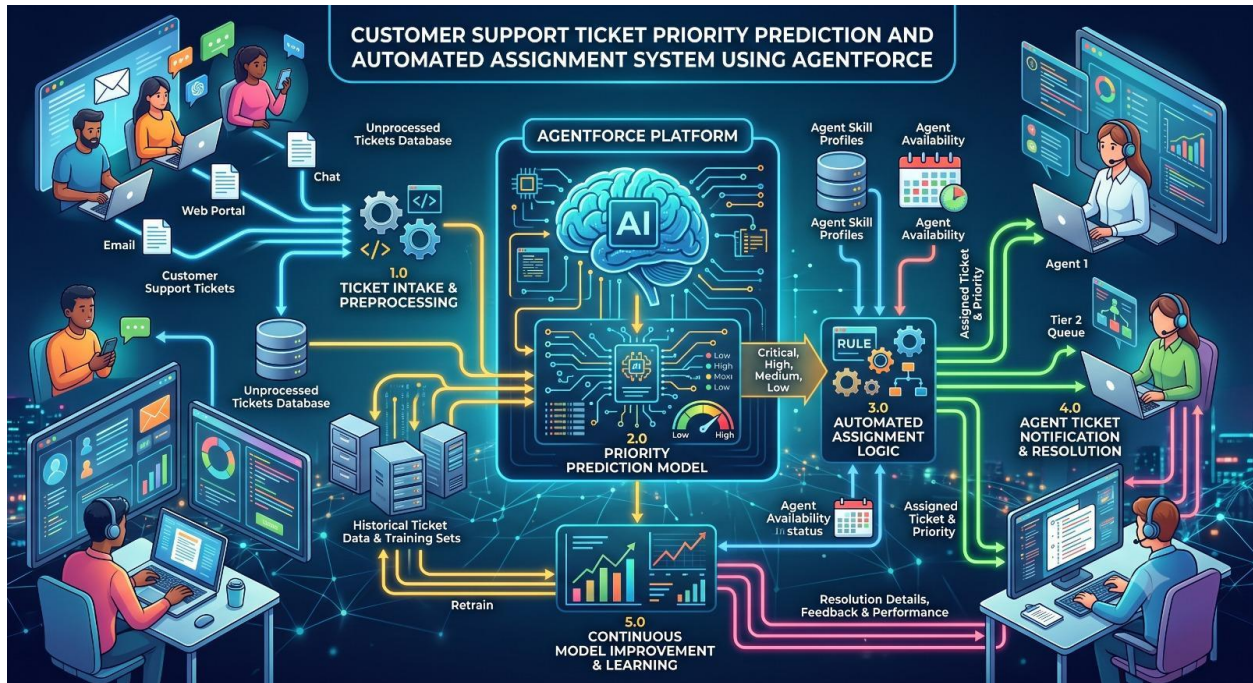
3.3 Non-Functional Requirements

- Usability: Clear AI messages
- Reliability: Consistent classification
- Performance: Quick analysis
- Security: Controlled Salesforce access
- Maintainability: Configurable flows
- Scalability: Handle growing ticket volumes

3.3 Data Flow Diagram

Flow: Customer / Support Request → Account → Latest Support Ticket → Ticket Description → Analyze Description → Priority Level → High / Medium / Low Decision → Agent Assignment → High-Priority Task (if applicable) → Final Response

Agentforce Flow: User provides *Account Name* → Agentforce Support Ticket Priority Analysis → Auto-Launched Flow → Ticket Analysis → Priority and Assignment Output → Response to User



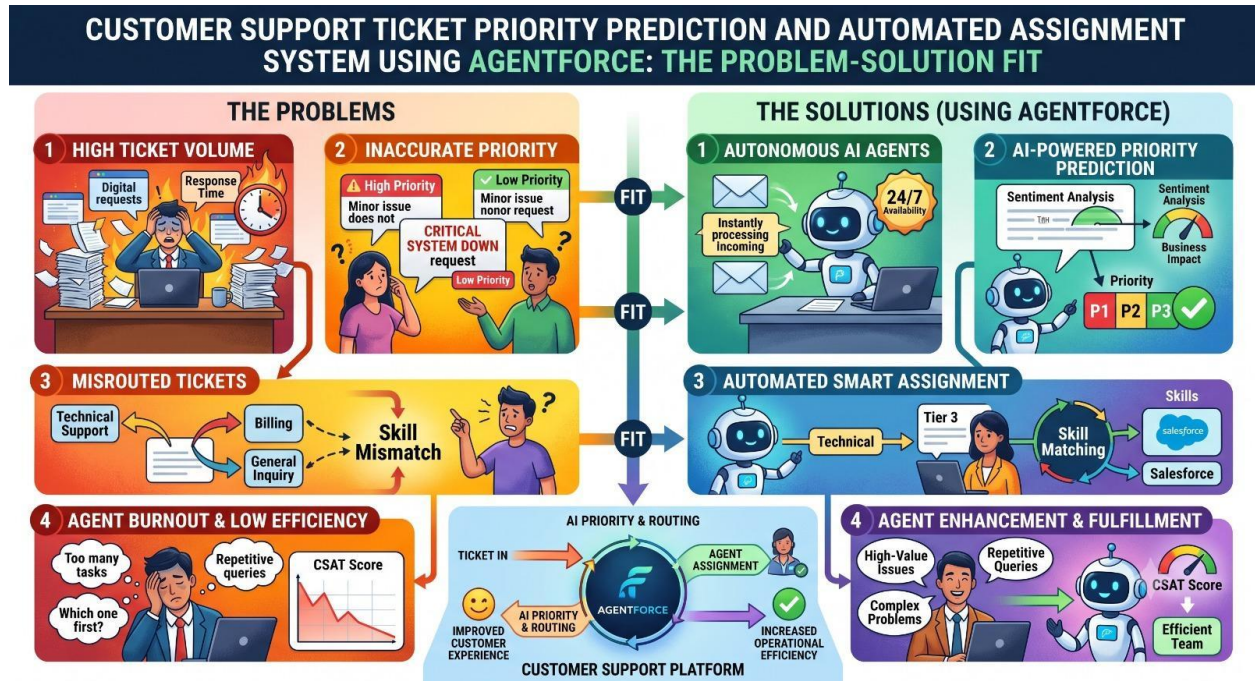
3.4 Technology Stack

Layer	Technology
Platform	Salesforce CRM
Database	Support_Ticket_Intelligence_c
Logic Layer	Auto-Launched Flow
AI Layer	Agentforce AI
Agent Configuration	Support Ticket Priority Analysis Subagent
Automation Layer	Flow Decision & Create Records
Output Layer	Agentforce Response

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Manual prioritization	AI-based classification	Faster detection of urgent cases
Missed urgency	Keyword-based rules	Critical issues escalated
Manual task creation	Automated urgent task	Reduced effort
Manual assignment	Flow-based assignment	Consistency
SLA risk overlooked	SLA monitoring	Better compliance



4.2 Proposed Solution

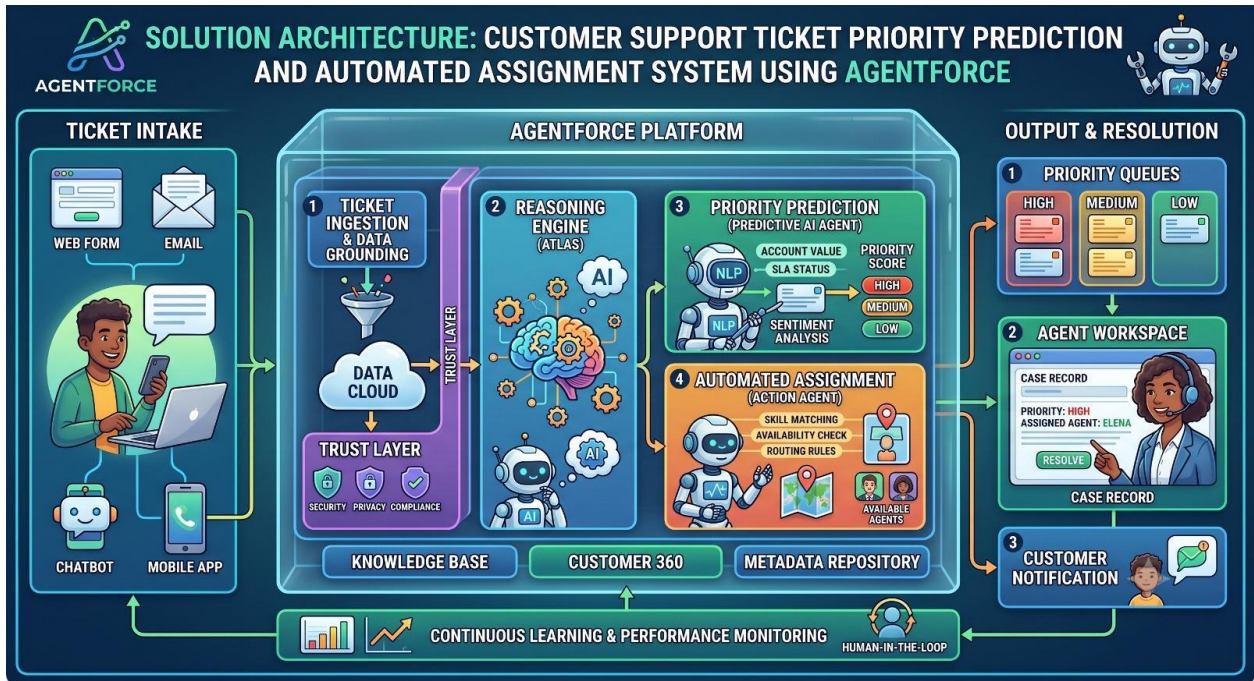
Functional Requirements

- **Problem:** Manual prioritization and assignment cause delays.
- **Solution:** Salesforce + Agentforce AI for automated classification and assignment.
- **Innovation:** NLP-based urgency detection, auto-launched flows, urgent task creation.
- **Impact:** Faster resolution, reduced manual effort, improved SLA compliance.
- **Scalability:** Extendable rules, dashboards, SLA monitoring.

4.3 Solution Architecture

Flow:

Customer → Agentforce → Ticket Priority Analysis → Auto-Launched Flow → Priority Decision → Task Creation → Assignment → Notifications → Resolution



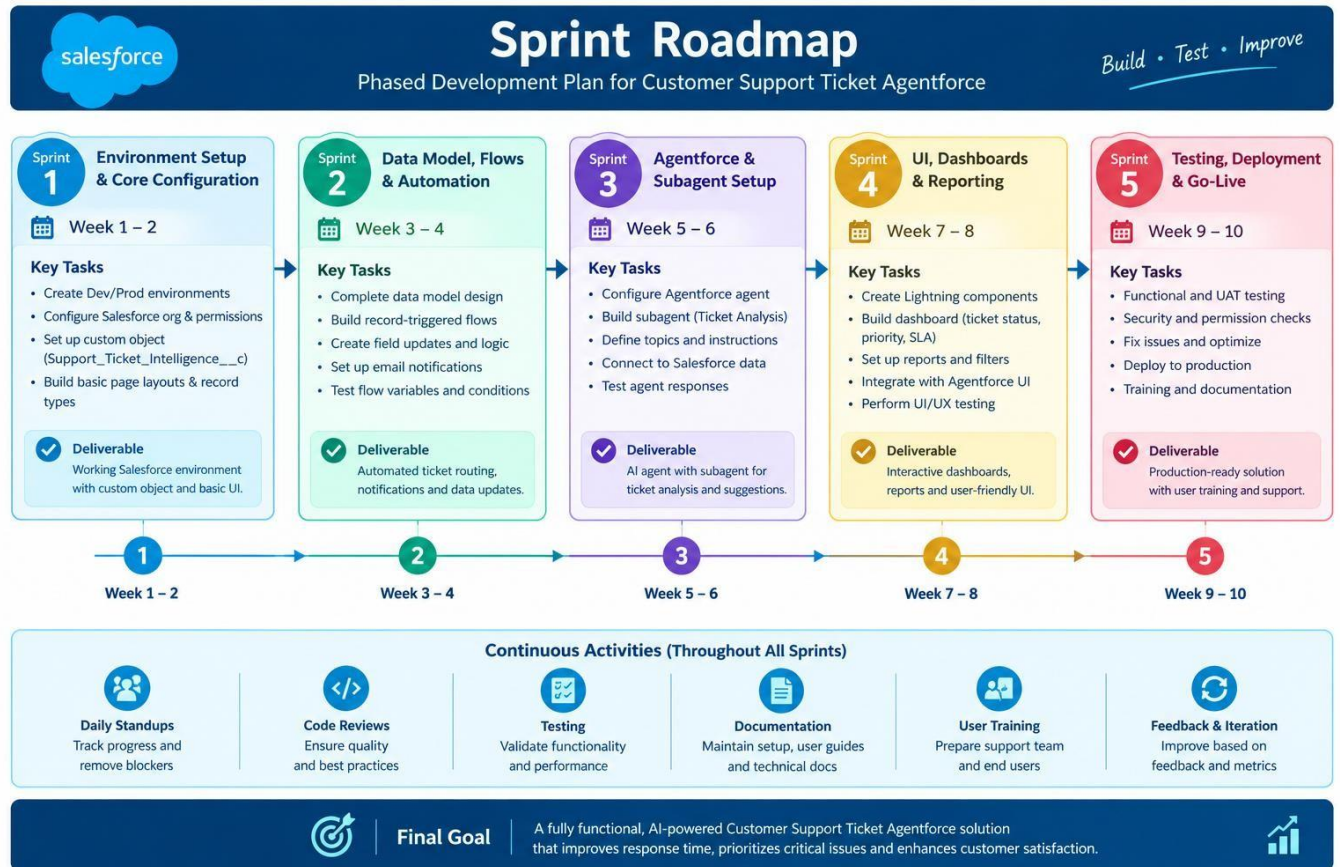
5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology with Epics → Stories → Sprints.
- Velocity tracking and burndown charts for progress.

5.2 Sample Sprint Backlog

- Sprint-1: Developer setup (Salesforce environment).
- Sprint-2: Data modeling (custom object creation).
- Sprint-3: Automation (flows for classification).
- Sprint-4: Agentforce integration.
- Sprint-5: SLA monitoring.
- Sprint-6: Conversational AI support.



6. Project Development Phase:

6.1 Learning Outcomes

- Data modeling in Salesforce
- Custom objects and relationships
- Lightning App Builder
- Record-triggered flows
- Apex triggers and batch jobs
- Scheduled Apex for SLA checks

Milestone 1: Developer Account Setup



Build for the Agentic Enterprise. Sign up for your Developer Edition.

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required
- ✓ Drag-and-drop when you want it, Apex when you need it
- ✓ 60+ MCP tools and 30+ coding skills, ready out of the box



Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

Country/region

India



State/province

Tamil Nadu



Work email

keerthanaravi2730@gmail.com



☐ I agree to the [Main Services Agreement – Developer Services](#), [Salesforce Program Agreement](#), and [API Terms of Service](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

We value your privacy. To learn more, visit our [Privacy Statement](#).

Sign me up

Worldwide

Legal

[Terms of Service](#)

[Privacy Information](#)

[Responsible Disclosure](#)

[Trust](#)

[Contact](#)

[Cookie Preferences](#)

[Your Privacy Choices](#)

© Copyright 2026 Salesforce, Inc. All rights reserved. Various trademarks held by their respective owners. Salesforce, Inc. Salesforce Tower, 415 Mission Street, 3rd Floor, San Francisco, CA 94105, United States



Build for the Agentic Enterprise. Sign up for your Developer Edition.

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required
- ✓ Drag-and-drop when you want it, Apex when you need it
- ✓ 60+ MCP tools and 30+ coding skills, ready out of the box



Almost there, keerthana! Check your email.

You'll receive a confirmation email soon. If you don't see the email, check your spam folder.

If we're getting a lot of signups at once, it can take a while for us to set up your Developer Edition. We'll send another email when it's ready.

Follow guided steps to finish setting it up, and use in-app guidance to learn Agentforce.

Worldwide ▼

[Legal](#) [Terms of Service](#) [Privacy Information](#) [Responsible Disclosure](#) [Trust](#) [Contact](#) [Cookie Preferences](#) [Your Privacy Choices](#)

© Copyright 2026 Salesforce, Inc. All rights reserved. Various trademarks held by their respective owners. Salesforce, Inc. Salesforce Tower, 415 Mission Street, 3rd Floor, San Francisco, CA 94105, United States



Ready for a new password?

Reset Password

Change Your Password

Enter a new password for

keerthanaravi2730.8de0d4e24d2a@agentforce.com.

Make sure to include at least:

- ✓ 8 characters
- ✓ 1 letter
- ✓ 1 number

* New Password

.....

Good

* Confirm New Password

.....

Match

* New Security Question

▼ In what city were you born?

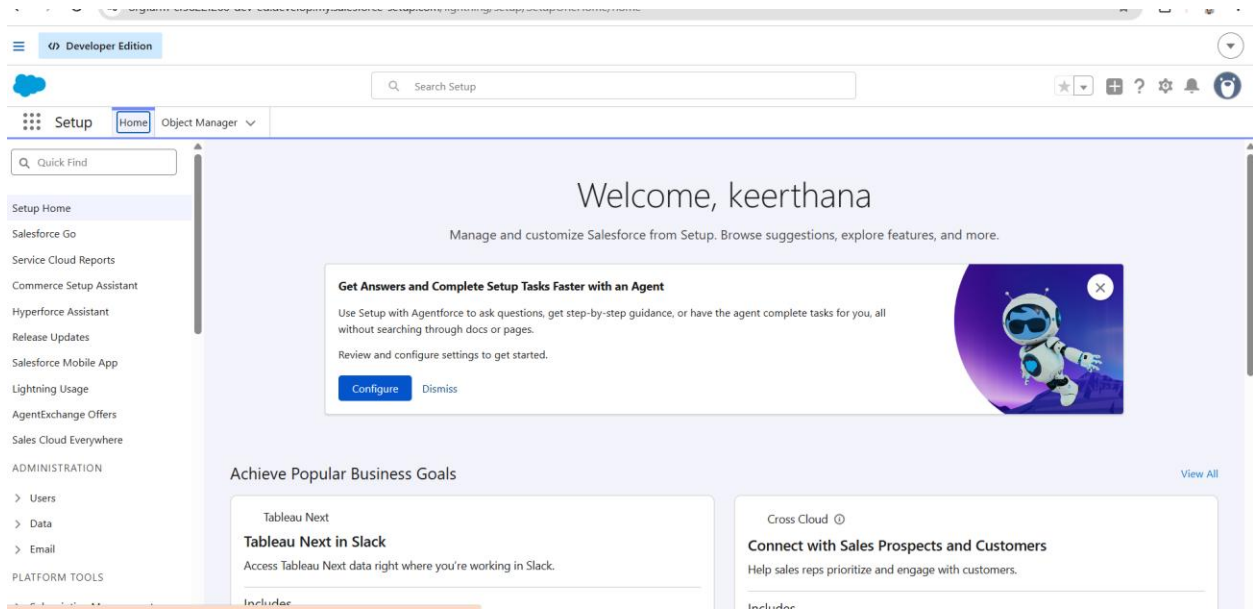
* New Answer

Ullikkottai

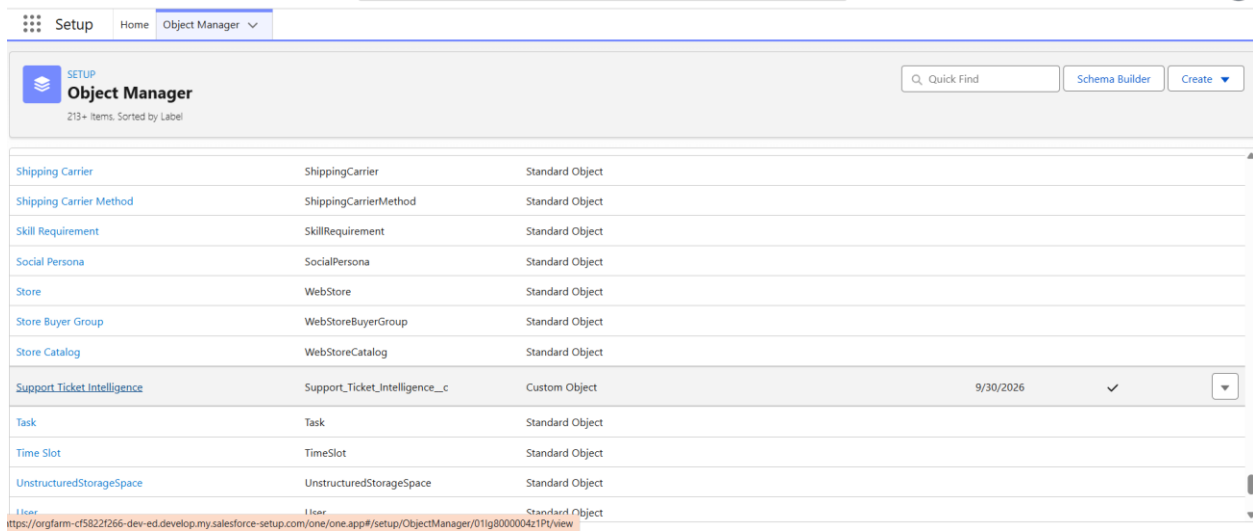
Good

*=required

Change Password



Milestone 2: Object Creation and Relationships



Developer Edition

Search Setup

Setup Home Object Manager

SETUP > OBJECT MANAGER

Support Ticket Intelligence

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Details

Description

API Name

Support_Ticket_Intelligence__c

Custom

✓

Singular Label

Support Ticket Intelligence

Plural Label

Support Ticket Intelligence

Enable Reports

Track Activities

Track Field History

Deployment Status

Deployed

Help Settings

Standard salesforce.com Help Window

EditDelete

Developer Edition

Search Setup

Setup Home Object Manager

Quick Find

Setup Home

Salesforce Go

Service Cloud Reports

Commerce Setup Assistant

Hyperforce Assistant

Release Updates

Salesforce Mobile App

Lightning Usage

AgentExchange Offers

Sales Cloud Everywhere

ADMINISTRATION

> Users

> Data

> Email

PLATFORM TOOLS

Recent Items

NAME	TYPE	OBJECT
keerthana ravichandran	User	
Resolution Time (hrs)	Custom Field Definition	Support Ticket Intelligence
SLA Breach Risk	Custom Field Definition	Support Ticket Intelligence
Assigned Resource	Custom Field Definition	Support Ticket Intelligence
Created Date	Custom Field Definition	Support Ticket Intelligence
Status	Custom Field Definition	Support Ticket Intelligence
Priority Level	Custom Field Definition	Support Ticket Intelligence
Description	Custom Field Definition	Support Ticket Intelligence
Issue Type	Custom Field Definition	Support Ticket Intelligence
Contact	Custom Field Definition	Support Ticket Intelligence

6.2 Auto-Launched Flow Logic (Modified for this Project)

Setup

Home

Object Manager

Quick Find

SETUP

Flows

Flow Trigger Explorer

New Flow

Flow Definitions

All Flows

103 items • Sorted by Flow Label • Filtered by All flow definitions • Updated a few seconds ago

⚙️

🔄

⌵

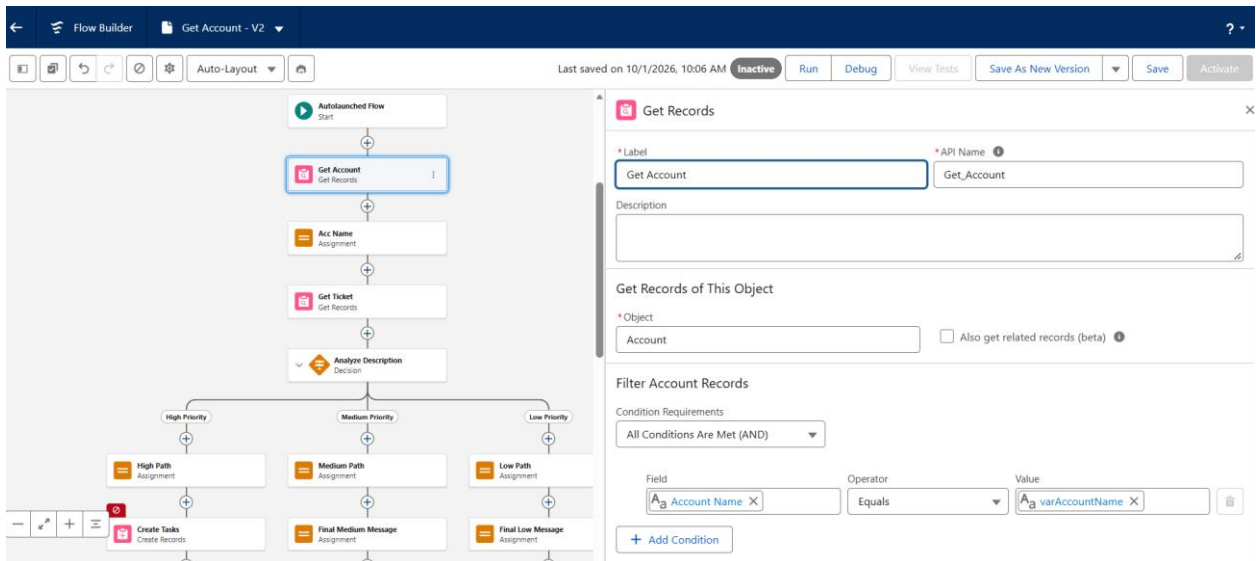
Flow Label	Process Type	Active	Template	Package State	Packag...	Last M...	Last M...
Send Verification Code	Autolaunched Flow	✓	☐	Managed-Installed			
Set Payment Rates	Salesforce Scheduler Flow	✓	✓	Managed-Installed			
Support_Ticket_Intelligence	Autolaunched Flow	✓	☐	Unmanaged		keerthana r...	9/30/2026, ...
Update CI	Autolaunched Flow	✓	☐	Managed-Installed			
Update Customer Contact Record	Autolaunched Flow	✓	☐	Managed-Installed			
Update Flow Definition Version Status	Autolaunched Flow	✓	✓	Managed-Installed			
Update Verified Contact	Autolaunched Flow	✓	☐	Managed-Installed			
Verify Code	Autolaunched Flow	✓	☐	Managed-Installed			
Verify Identity	Screen Flow	✓	✓	Managed-Installed			
Voice Calls Routed to Agents and Queues	Omni-Channel Flow	✓	✓	Managed-Installed			

3. Step-by-Step Flow Implementation

Step 1: Get Account

Element: Get Records

- **Label:** Get Account
- **Object:** Account
- **Condition:** ○ Name Equals {!varAccountName}
- **Sort:** CreatedDate DESC
- **Store:** Only first record



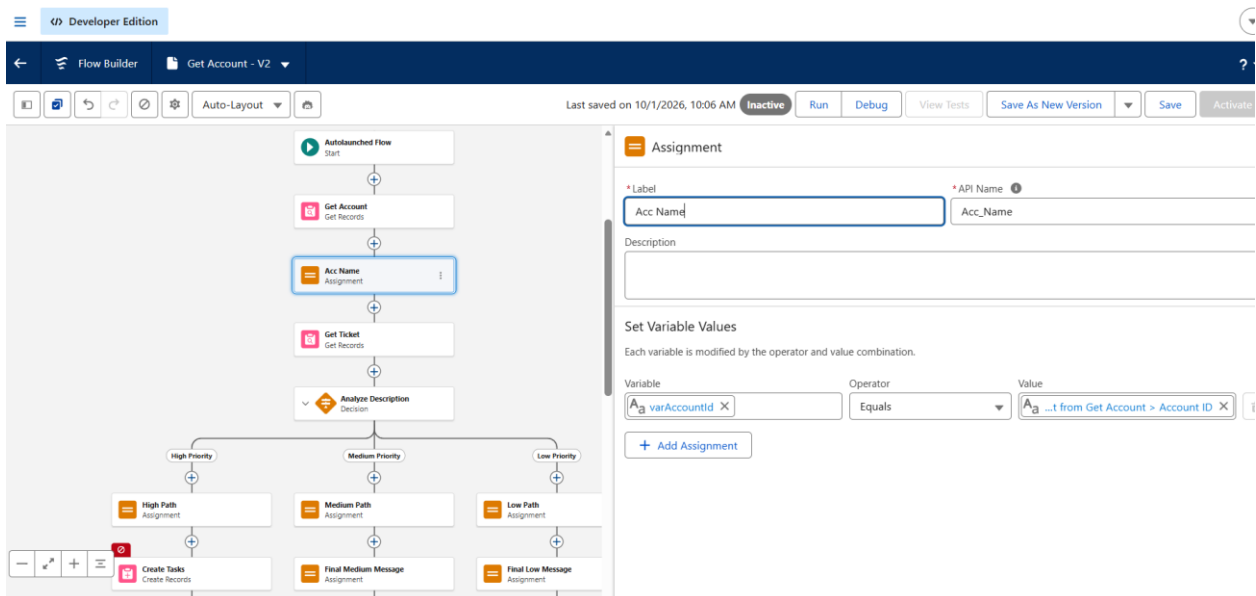
Step 2: Assignment (Store Account Id)

Element: Assignment

- **Label:** Store Account Id

Set:

- `varAccountId = {!Get_Account.Id}`



Step 3: Get Ticket Record Element:

Get Records

- Label: Get Ticket
- Object: Support_Ticket_Intelligence__c
- Condition:
 - Customer__c Equals {!varAccountId}
- Sort: CreatedDate DESC
- Store: Only first record

The screenshot displays the Salesforce Flow Builder interface for a flow named 'Get Account - V2'. The flowchart on the left shows the sequence of steps: 'Autolaunched Flow Start' leads to 'Get Account' (Get Records), then 'Acc Name Assignment', followed by 'Get Ticket' (Get Records), and then an 'Analyze Description' (Decision) step. The decision step branches into three paths: 'High Priority', 'Medium Priority', and 'Low Priority'. Each path leads to an 'Assignment' step ('High Path Assignment', 'Medium Path Assignment', and 'Low Path Assignment' respectively), followed by a 'Create Task' (Create Records) step and a 'Final' message step ('Final High Message Assignment', 'Final Medium Message Assignment', and 'Final Low Message Assignment' respectively).

The right-hand pane shows the configuration for the 'Get Records' step. The 'Label' is 'Get Ticket' and the 'API Name' is 'Get_Ticket'. The 'Object' is 'Support Ticket Intelligence'. The 'Condition Requirements' are set to 'All Conditions Are Met (AND)'. A single condition is added: Field 'Customer' (with a dropdown arrow), Operator 'Equals', and Value 'A ...m Get Account > Account ID' (with a dropdown arrow and a close button).

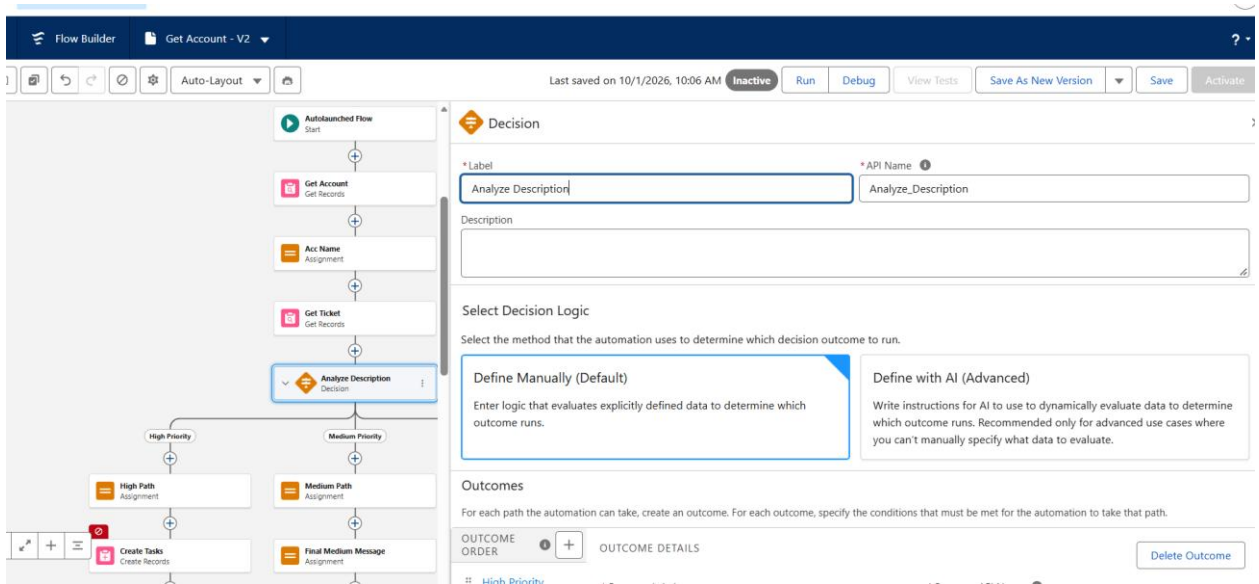
Step 4: Assignment (Store Ticket Id)

Element: Assignment

- Label: Store Ticket Id

Set:

- varTicketId = {!Get_Ticket.Id}



Step 5: Decision (Analyze Description)

Element: Decision

- **Label:** Analyze Description

Outcomes:

High Priority

- **Description contains:**

- "urgent"
- "not working"
- "failure"

Medium Priority

- **Description contains:**

- "issue"
- "slow"
- "delay"

Default Outcome

- Low Priority

Decision

✕

Outcomes

For each path the automation can take, create an outcome. For each outcome, specify the conditions that must be met for the automation to take that path.

OUTCOME ORDER

+

OUTCOME DETAILS

Delete Outcome

High Priority

Medium Priority

Low Priority

* Outcome Label

High Priority

* Outcome API Name

High_Priority

Condition Requirements to Execute Outcome

Any Condition Is Met (OR)

* Resource

Aa ...rom Get Ticket > Description

Operator

Equals

Value

urgent

✕

OR

* Resource

Aa ...rom Get Ticket > Description

Operator

Equals

Value

not working

✕

OR

* Resource

Aa ...rom Get Ticket > Description

Operator

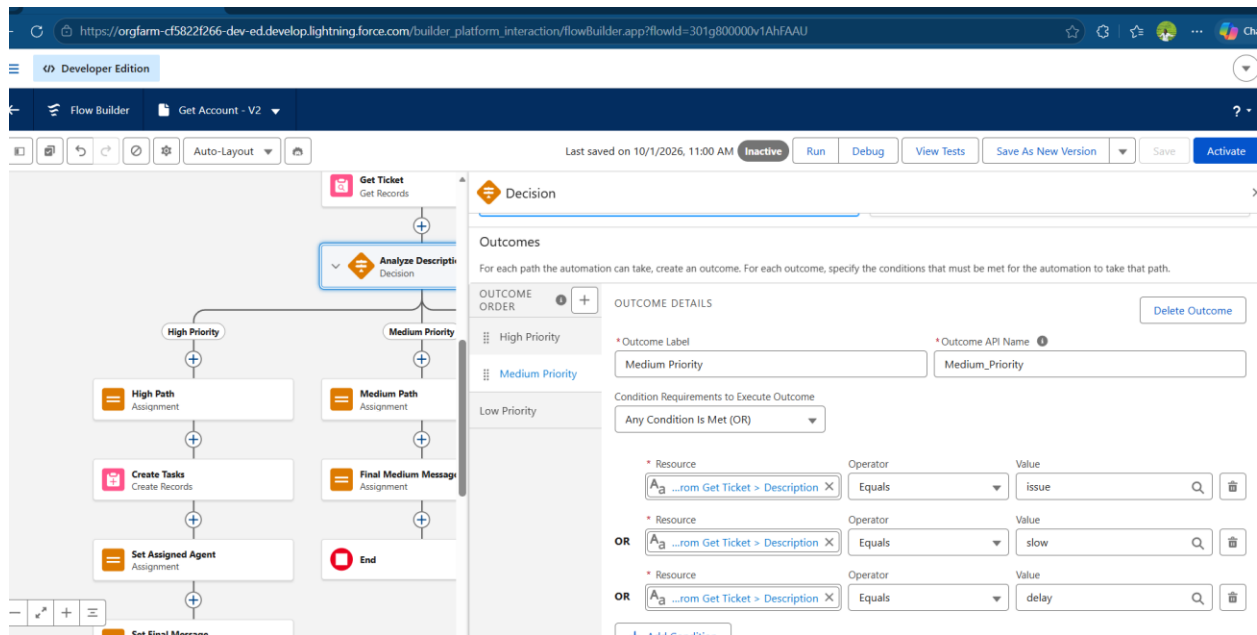
Equals

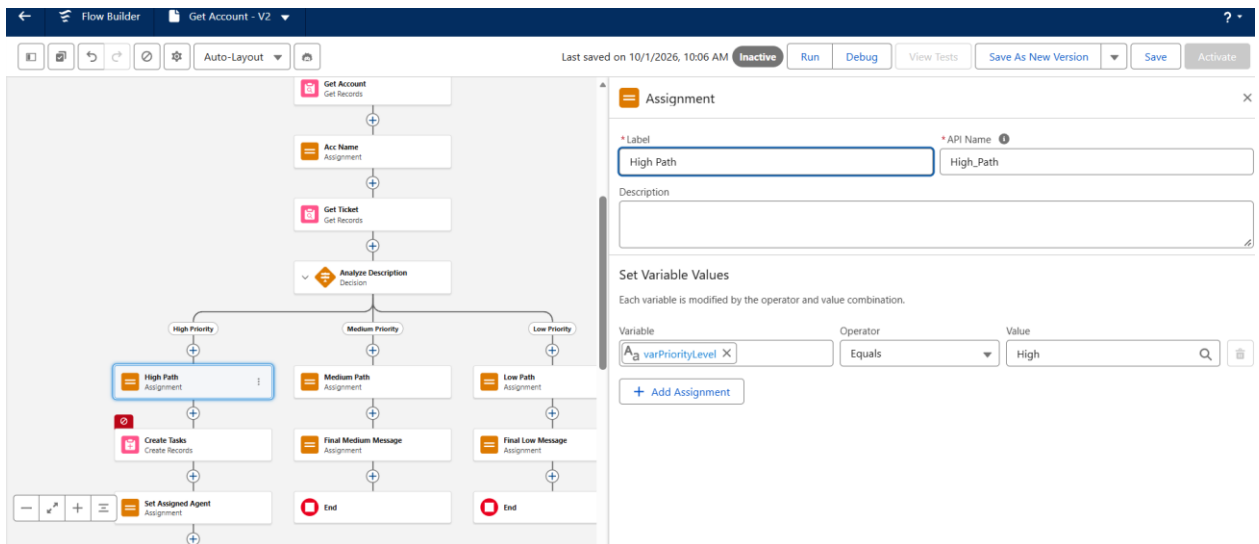
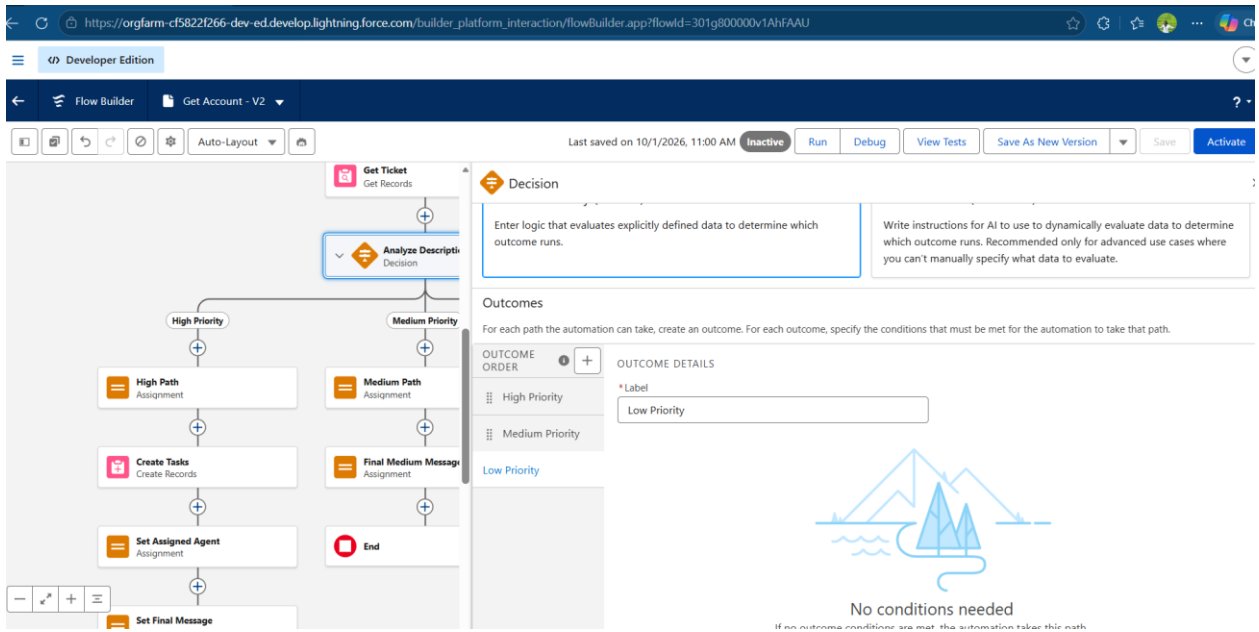
Value

failure

✕

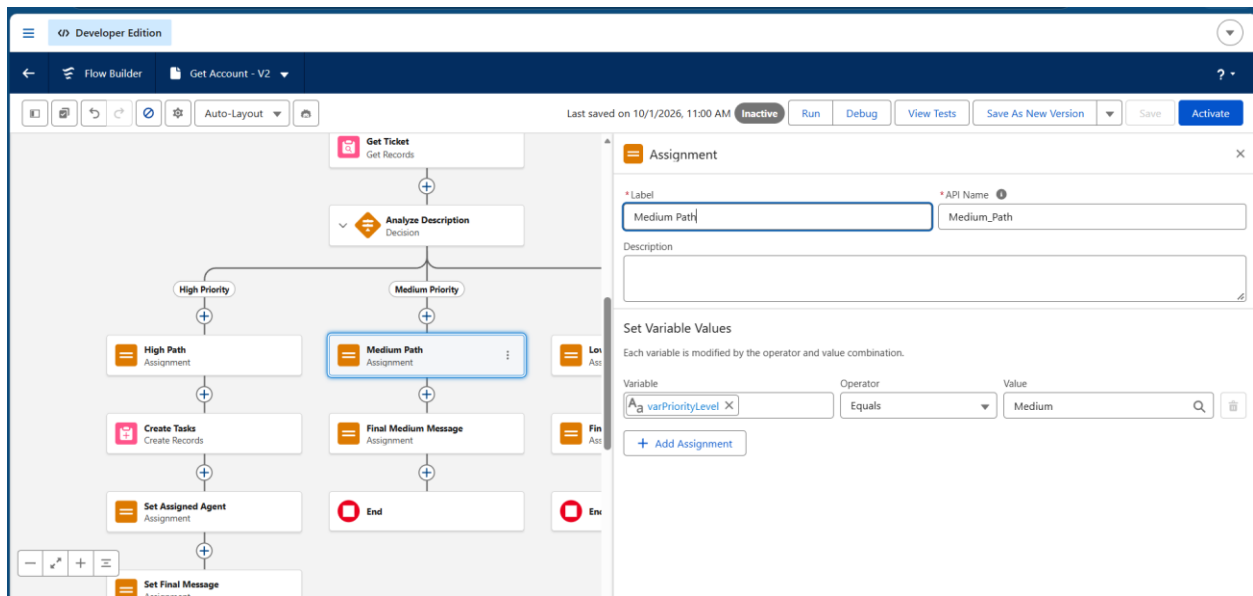
+ Add Condition





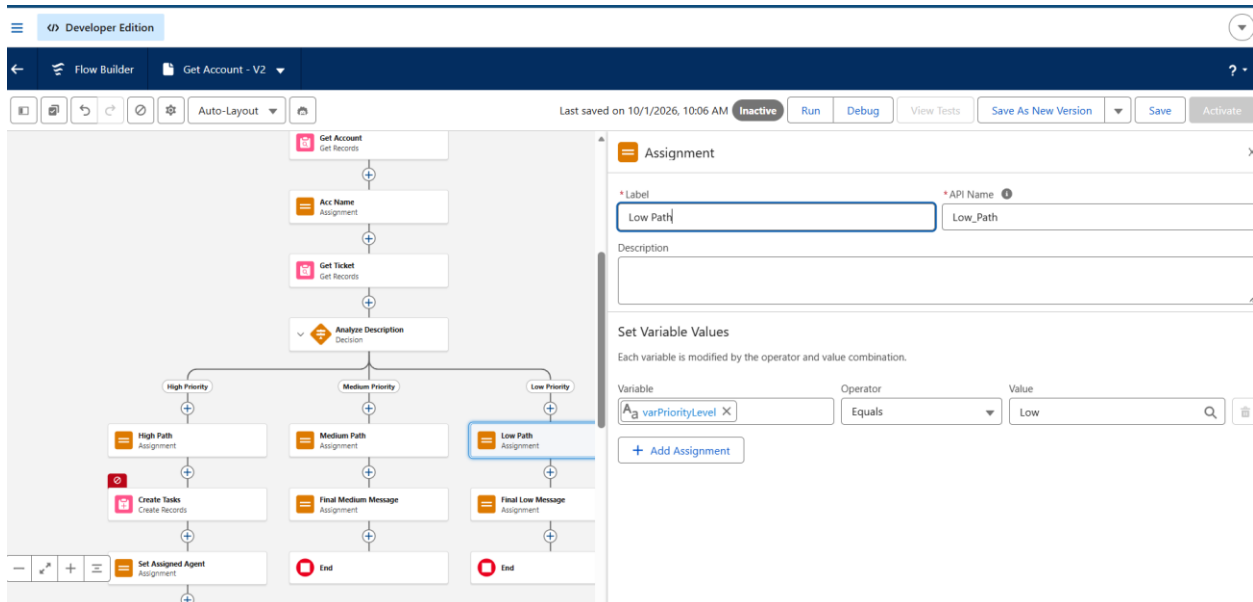
Medium Path

- `varPriorityLevel = "Medium"`



Low Path

- `varPriorityLevel = "Low"`



Step 7: Decision (High Priority Check)

Element: Decision

- Label:
Is High Priority

Condition:

- varPriorityLevel Equals "High"

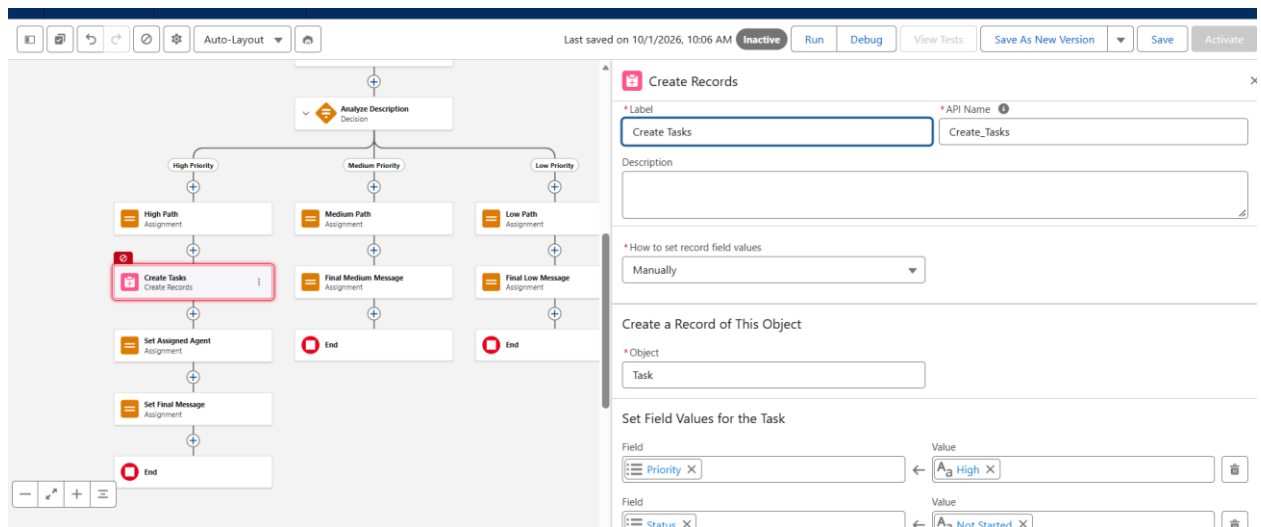
Step 8: Create Task (Only High Priority)

Element: Create Records

- Object: Task Set

Fields:

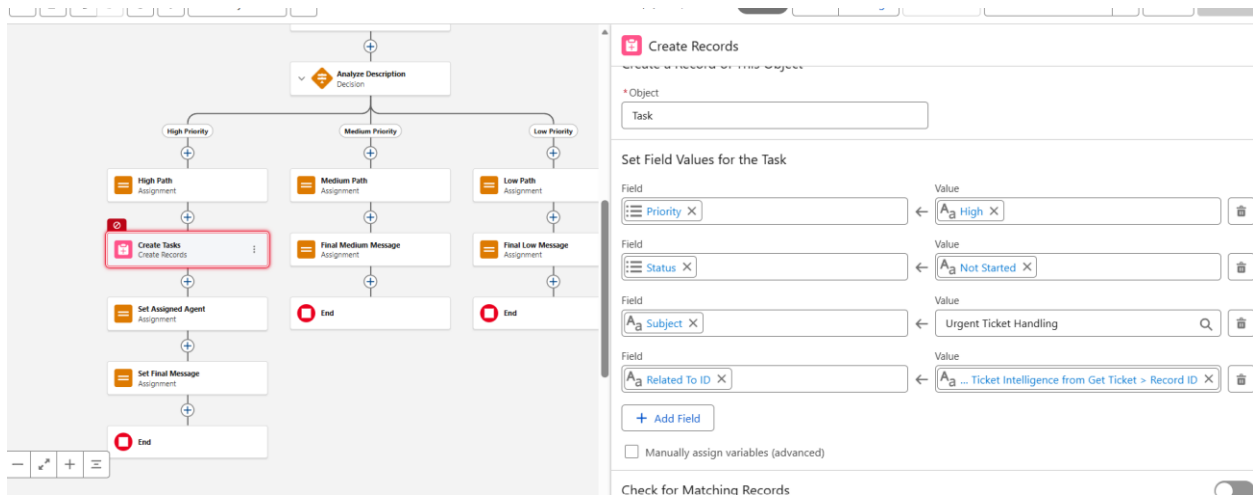
- Subject = "Urgent Ticket Handling"
- WhatId = {!varTicketId}
- Priority = High
- Status = Not Started



The screenshot displays a workflow editor interface. At the top, a toolbar includes icons for undo, redo, and other actions, along with a status bar indicating 'Last saved on 10/1/2026, 10:06 AM' and buttons for 'Inactive', 'Run', 'Debug', 'View Tests', 'Save As New Version', 'Save', and 'Activate'.

The main workspace shows a decision node labeled 'Analyze Description' with three outgoing paths: 'High Priority', 'Medium Priority', and 'Low Priority'. The 'High Priority' path is highlighted with a red box and contains a 'Create Tasks' node, which is also highlighted with a red box. Below this node are 'Set Assigned Agent' and 'Set Final Message' nodes, followed by an 'End' node. The 'Medium Priority' path contains a 'Final Medium Message' node and an 'End' node. The 'Low Priority' path contains a 'Final Low Message' node and an 'End' node.

The right sidebar shows the configuration for the 'Create Records' element. It includes fields for 'Label' (Create Tasks) and 'API Name' (Create_Tasks). The 'Description' field is empty. Below this, a dropdown menu for 'How to set record field values' is set to 'Manually'. The 'Create a Record of This Object' section shows the 'Object' field set to 'Task'. The 'Set Field Values for the Task' section shows two fields: 'Priority' with a value of 'High' and 'Status' with a value of 'Not Started'.



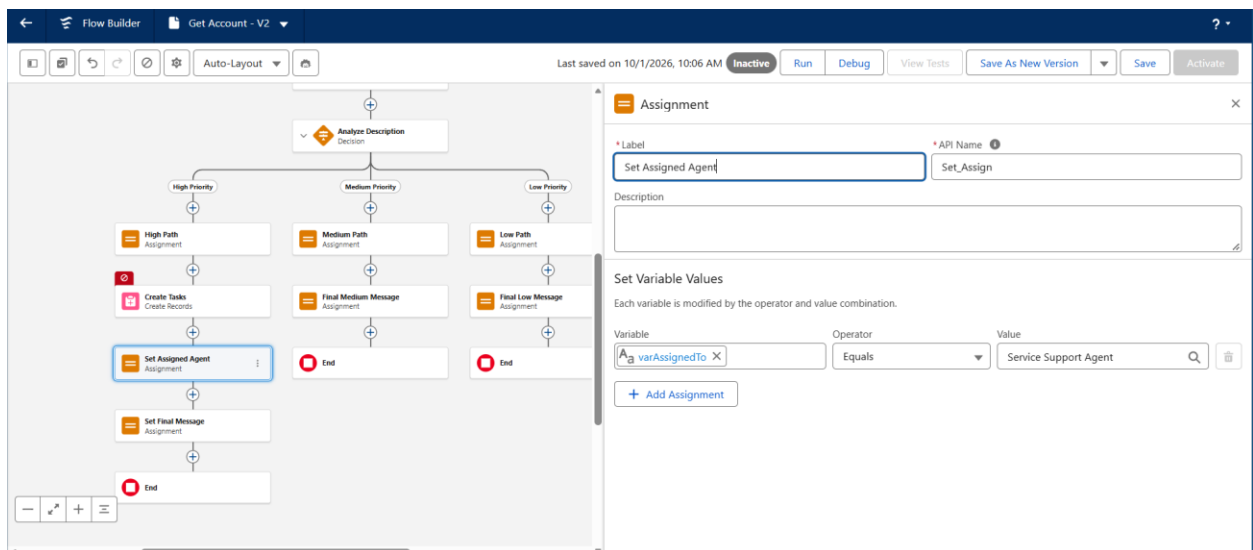
Step 9: Assignment (Assign Agent Name)

Element: Assignment

- Label: Set Assigned Agent

Set:

- varAssignedTo = "Senior Support Agent"



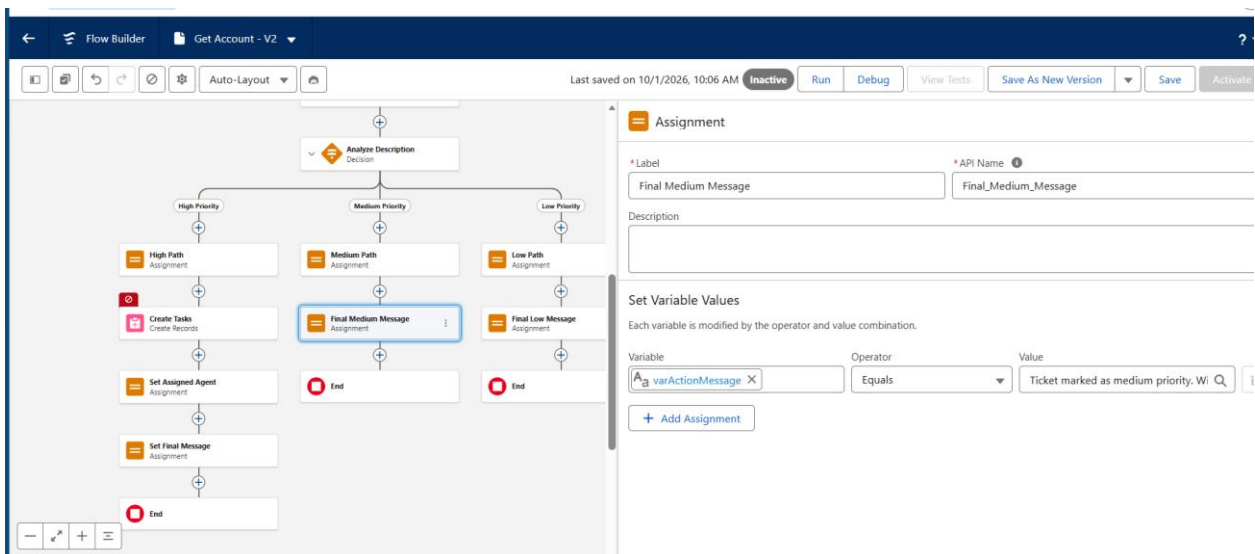
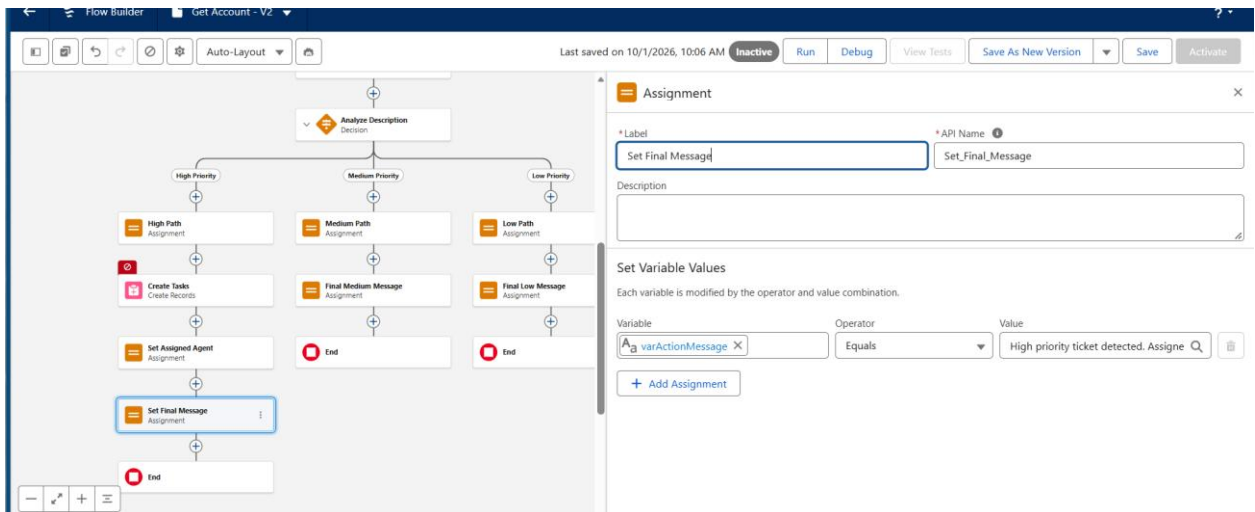
Step 10: Final Output Message

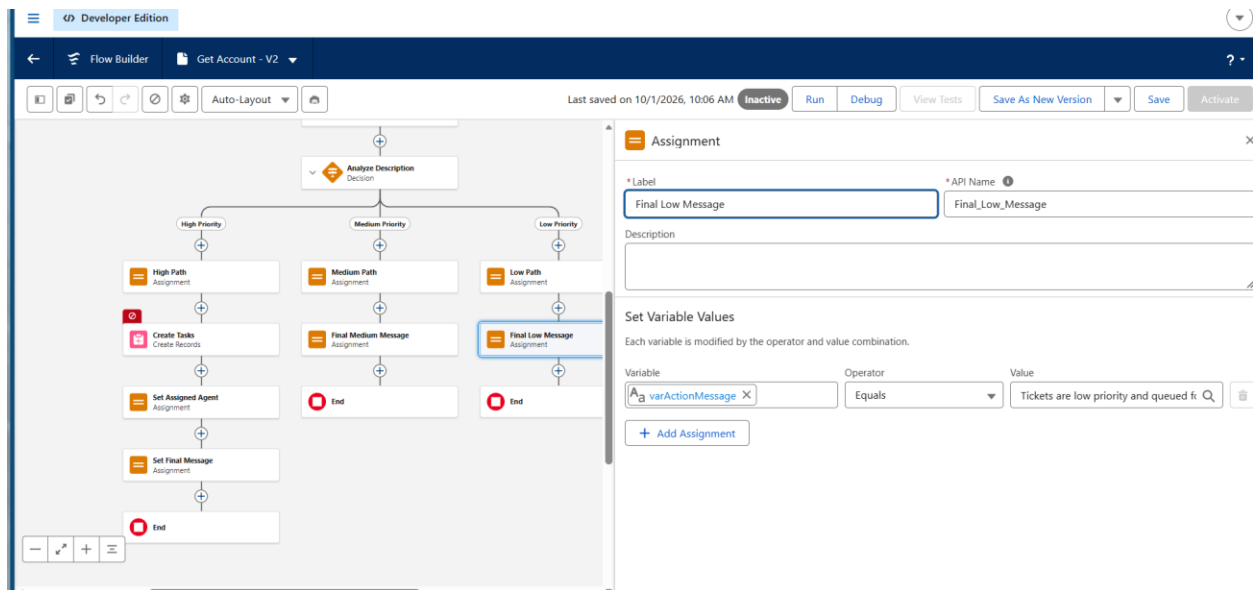
Element: Assignment

- Label: Set Final Message

Set varActionMessage:

- High: "High priority ticket detected. Assigned to senior agent."
- Medium: "Ticket marked as medium priority. Will be handled shortly."
- Low: "Tickets are low priority and queued for processing."





Step 11 : Save Flow Label

Name : Support_Ticket_Intelligence

- Active Flow

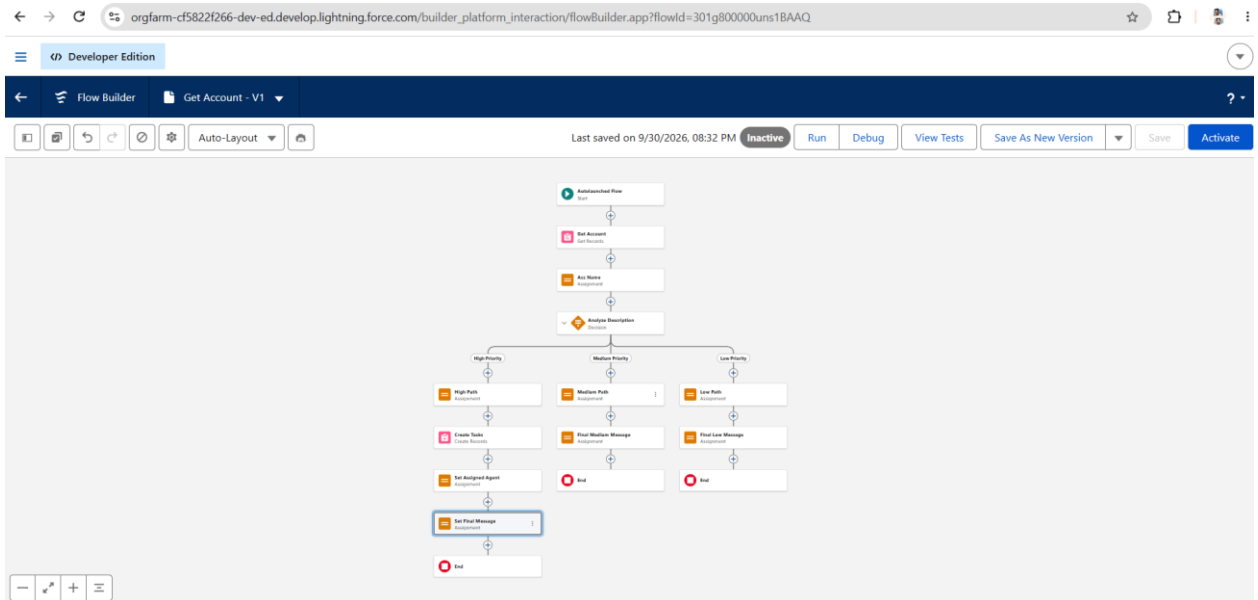
Final Flow Structure

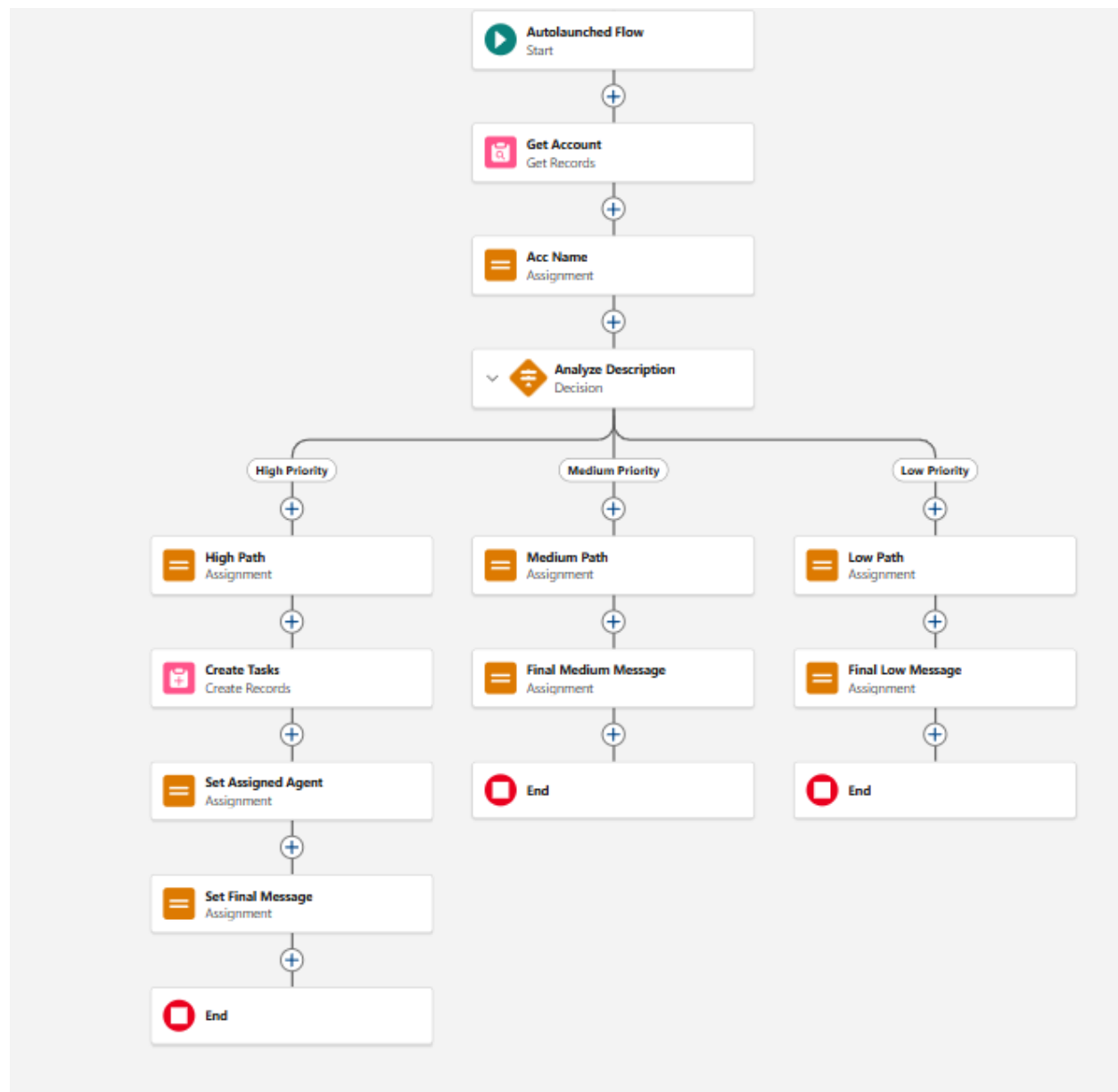
Start

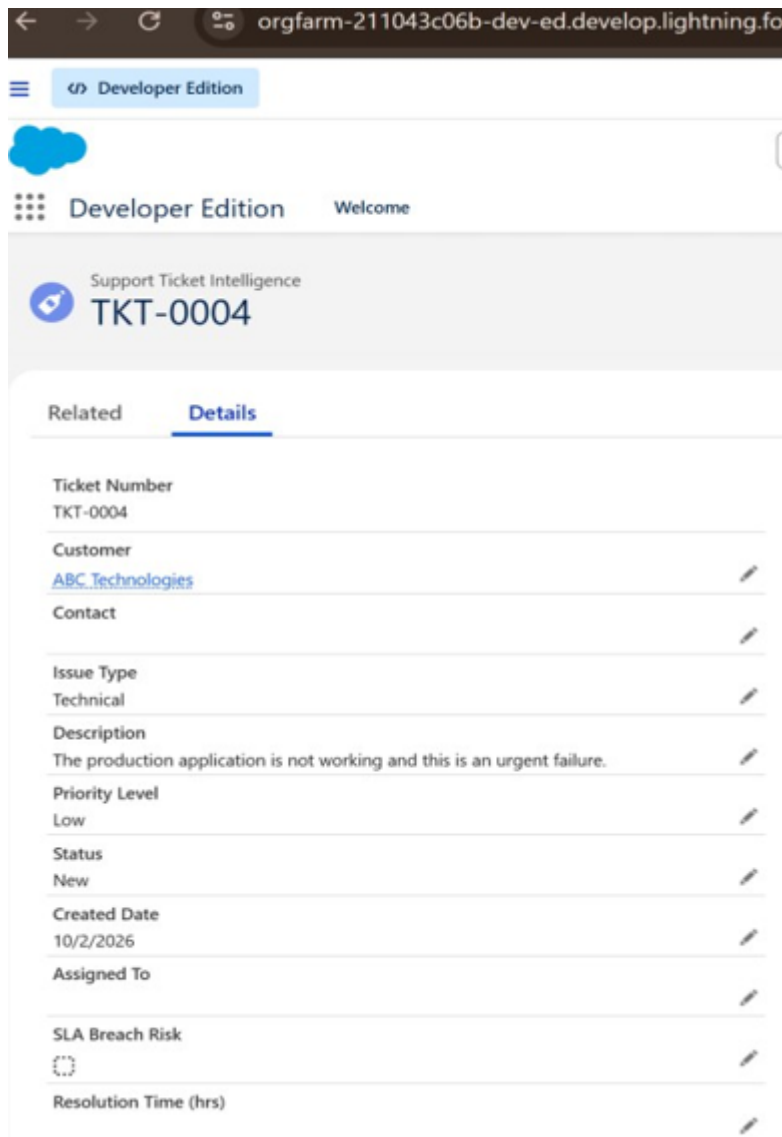
- Get Account
- Assignment (Account Id)
- Get Ticket
- Assignment (Ticket Id)
- Decision (Analyze Description)
- Assignment (Priority Level)
- Decision (High Priority)
- Create Task
- Assignment (Assigned Agent)
- Optional SLA Decision
- Assignment (Final Message)

→ End

Flow OutPut :







7. FUNCTIONAL & PERFORMANCE TESTING

The system uses Auto-Launched Flows in Salesforce to automate ticket handling without manual triggers. When a customer submits a request, the flow retrieves the latest ticket linked to the account, analyzes the description, and classifies it into High, Medium, or Low priority. Decision elements check for keywords such as “urgent,” “failure,” or “delay” to determine severity.

8. ADVANTAGES & DISADVANTAGES

ADVANTAGES

☐ Improved SLA Compliance – Ensures urgent tickets are escalated quickly, minimizing breach risks.

☐ Error Reduction – Eliminates human mistakes in triaging and misrouting tickets.

☐ Scalability – Handles sudden spikes in ticket volume without overwhelming support teams.

☐ Customer Satisfaction – Provides faster responses and transparent updates to customers.

☐ Managerial Insights – Dashboards and reports give managers real-time visibility into performance.

☐ Continuous Learning – AI models improve over time with feedback, making predictions more accurate.

☐ Resource Optimization – Assigns tickets based on agent skills and workload, balancing efficiency.

DIS ADVANTAGES

☐ Implementation Complexity – Requires careful setup of Salesforce objects, flows, and AI integration.

☐ Initial Costs – Investment in Salesforce licenses, Agentforce configuration, and training can be high.

☐ Dependency on AI Accuracy – Misclassification of tickets may still occur if the model is not well-trained.

- Customization Effort – Flows and routing rules must be tailored to each organization's processes.
- Maintenance Overhead – Continuous monitoring and updates are needed to keep automation effective.
- Limited Human Judgment – AI may miss nuanced issues that experienced agents would catch.
- Data Security Concerns – Sensitive customer data must be protected during AI analysis and routing.

9. CONCLUSION:

The Customer Support Ticket Priority Prediction and Automated Assignment System successfully demonstrates how Salesforce CRM and Agent force AI can transform traditional support workflows. By automating ticket prioritization, reducing manual errors, and ensuring SLA compliance, the system enhances both agent efficiency and customer satisfaction.

10. FUTURE SCOPE

- Add more priority indicators and business rules.
- Extend Agent force outputs with additional ticket information.
- Add more sophisticated SLA monitoring and escalation.
- Support additional support teams or assignment rules.
- Introduce analytics for priority trends and resolution performance.

11. APPENDIX

- Custom Object: Support Ticket Intelligence
- Key Salesforce Records: Account, Contact, User, Task
- Flow: Support_Ticket_Intelligence – Auto-Launched Flow
- Agentforce Subagent: Support Ticket Priority Analysis
- Agentforce Action: Flow-based Support Ticket Intelligence action

THANK YOU